



Maximum Matchings

Lecture 9 | Algorithms Design & Analysis

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Idea: Augmenting Paths



Definition 3. If G is a graph and M is a matching in G , a vertex is called *matched* if it belongs to one of the edges in M , and *free* otherwise.

An *alternating component with respect to M* (also called an *M -alternating component*) is an edge set that forms a connected subgraph of G of maximum degree 2 (i.e., a path or cycle), in which every degree-2 vertex belongs to exactly one edge of M . An *augmenting path with respect to M* is an M -alternating component which is a path both of whose endpoints are free vertices.

Idea: Augmenting Paths



In the following lemma, and throughout these notes, we use the notation $A \oplus B$ to denote the *symmetric difference* of two sets A and B , i.e. the set of all elements that belong to one of the sets but not the other.

Lemma 1. *If M is a matching and P is an augmenting path with respect to M , then $M \oplus P$ is a matching containing one more edge than M .*

Lemma 2. *A matching M in a graph G is a maximum cardinality matching if and only if it has no augmenting path.*

Algorithm 1 Naïve iterative scheme for computing a maximum matching

- 1: Initialize $M = \emptyset$.
 - 2: **repeat**
 - 3: Find an augmenting path P with respect to M .
 - 4: $M \leftarrow M \oplus P$
 - 5: **until** there is no augmenting with respect to M .
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How to find Augmenting Paths?



Definition 4. If $G = (L, R, E)$ is a bipartite graph and M is a matching, the graph G_M is the directed graph formed from G by orienting each edge from L to R if it does not belong to M , and from R to L otherwise.

Lemma 3. Suppose M is a matching in a bipartite graph G , and let F denote the set of free vertices. M -augmenting paths are in one-to-one correspondence with directed paths from $L \cap F$ to $R \cap F$ in G_M .

Read



Section 1 in

<https://www.cs.cornell.edu/courses/cs6820/2023fa/handouts/matchings.pdf>

Section 1.1 in

<https://math.mit.edu/~goemans/18433S09/matching-notes.pdf>

Solve



Exercises 1.1 - 1.6 in

<https://math.mit.edu/~goemans/18433S09/matching-notes.pdf>